

TASK RISK ASSESSMENT (TRA)

Project (If Applicable)		TRA Ref. No.	DOUS-TRA-002	
Location / Worksite				
Sub Location / Area				
Equipment Spread (Optional)				
Task / Activity	LAUNCH AND RECOVERY			
Approver				
Responsible Persons				
Date TRA Created				
Reference Supporting Documents & Task Plan				
S3RA:	Hot Work Checklist:			
Task Plan:	Work at Height Checklist:			
Lift Plan:	Confined Space Work Checklist:			
PTW:	Electrical Equipment Checklist:			
COSHH / SDS:	Electrical Isolation Checklist:			
Rescue Plan:	Machinery Isolation Checklist:			
Work Equipment Assessment:	Pressure Systems Checklist:			
Other:	Other: DSS-RV-WK1-005 & DSS-RV-PRO-004			
Reason for TRA (Tick where Applicable)				
Identification of New Task	Annual Review	Review following Incident	Management of Change Record	Other, Please Specify
☐	☐	☐	☐	☐
Other (Details):				
TRA Participants				
Name	Position		Signature	
NOTE: STANDARD PPE OF SAFETY FOOTWEAR, SAFETY HELMET, COVERALLS, GLASSES AND GLOVES MUST BE WORN. ANY ADDITIONAL PPE REQUIRED SHALL BE CAPTURED IN THE MITIGATING CONTROLS SECTION OF THE TRA TEMPLATE BELOW.				

No .		Task /Activity	Hazard and Consequence		Initial Risk			Mitigating Controls	Residual Risk		
		Sequential Steps	Hazard	Consequence	Probability	Consequence Severity	Risk Rating	Measures to Reduce Risk to ALARP	Probability	Consequence Severity	Risk Rating
		ROV LAUNCH									
1		Review TRA and TBT Review DSS-RV-WK1-005 & DSS-RV-PRO-004	Miscommunication	Personnel Injury / Environmental Pollution	1	2	M	Situational awareness and proper use of PPE. Hard hats, impact gloves, cotton gloves, latex gloves, steel toed boots, safety glasses, side shields and face shield	1	1	L
2		Prior to any task being undertaken ensure that all personnel are aware of the hazards involved.	Electrical shock, possible slips trips and falls. Pinch points poor body position. Hydraulic discharge and spills.	Personnel Injury / Environmental Pollution	2	3	M	All personnel must be aware of their surrounds and weather conditions. Use proper Manual Handling techniques, i.e., Always use proper body positioning, lift with legs, and obtain second personnel when loads are over 40lbs. Read all SDS on chemicals used, use PPE that is specified in SDS and be aware of where all spill kits are located.	2	2	L
3		Pilot, Winch Operator, Recovery/Launch Observer all on comms	No communication between personnel causing injury to personnel and damage to equipment	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	L	All personnel involved must remain in constant communication	1	1	L
4		Check weather condition prior ROV leaving the deck and obtain green light from the Bridge personnel, OCM and Client	Damage to equipment from weather patterns or vessel movement	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Communication with bridge personnel, OCM and Client prior to lifting system from deck.	1	1	L
5		All crew members must perform a safety sweep and check for tools, and other loose items on the TMS and ROV before launch.	Marine debris	Environmental Pollution	2	2	L	Crew members must be aware of all tools and other items that were used during maintenance and be sure that all items have been put away in proper locations	1	1	L

Task Risk Assessment Form

6		Energize Winch HPU.	Damage to equipment.	Personnel Injury / Environmental Pollution / Damage to Equipment, ensure cooling fan start on HPU start up	3	4	M	Winch operator obtains visual of lights and listens to HPU upon energizing system and obtain visual of all gauges and ensure that system is at proper operating pressures before leaving HPU and check all gauges.	2	2	L
7		Ensure cooling water pump is on. If system has one. If HPU contains air cooled system, ensure that fan is operational.	Damage to HPU from overheating	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Launch/Observer confirms cooling water is on and flowing with visual of water going overboard.	2	2	L
8		Recovery/Launch Observer will remove grounding strap and cargo straps from ROV.	Possible electrical shock when grounding strap has been removed. Straps under tension	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Do not touch the ROV after grounding strap has been removed. Place cargo straps in secure area out of walkways.	2	2	L
9		Opening of LARS gates. Recovery/Launch Observer is to be 100% tied off with a full body harness with floatation.	. Falling overboard, injury to personnel	Personnel Injury	2	2	L	100% tie off to prevent the hazard of falling overboard, and or injury. <i>Crew member is NOT to open Gates if ROV has left the deck.</i>	1	1	L
10		Ensure that the spotter is visual inspecting the hydraulic hoses not being pinched in the hard stops	Hoses being crushed on hard stops. Impact,	Loss of containment, Injury to personnel, damage to equipment, injury to personnel	2	1	L	Spotter maintains visual of hydraulic hoses not being crushed in hard stops, ensure hoses move freely and DO NOT BIND			
11		Ensure that LARS is clear of Crane during Launch of ROV.	Contact with crane, damage to equipment and or personnel.	Personnel Injury / Damage to Equipment	3	4	M	Ensure that personnel maintain visual of LARS and Crane during launch of ROV.	2	2	L
12		Begin launching the ROV following the site-specific launch procedures.	Damage to equipment from failure to follow procedures	Personnel Injury / Damage to Equipment	3	4	M	Ensure personnel operating launch system is well trained and understands the system.	2	2	L
13		Closing LARS gates if gates have been opened.	Falling overboard, injury to personnel, line of fire	Personnel Injury	2	2	L	Crew member must have full body harness with floatation and 100% tie off to prevent the	1	1	L

								hazard of falling overboard, and or injury. <i>Crew member is NOT to Close Gates if ROV is out of Docking head and only if tension has been relieved of umbilical.</i>			
14		Unlatching from Docking head	Damage to Equipment	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator will switch from float to dive on docking head and will not exceed SWL of umbilical while pulling up to unlock latches. Monitor latch indicators and do not lower until 4 green indicators are fully extended.	2	2	L
15		ROV moving into and through splash zone.	Damage to ROV, separation of umbilical from shock load	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Controlled movement towards interface, timing wave patterns for entry into water. Winch operator will stop between 100 and 500FSW and switch from low to high tension, drive will be switched to float on docking head and continue to ascend per supervisor's request.	2	2	L
16		Constant communication between pilot and winch operator during descent.	Possible problems with ROV during descent, hitting seabed or structures. Failure of internal winch components, Winch Internal brake system and or Control pay in/out joystick.	Personnel Injury / Environmental Pollution / Damage to Equipment	4	5	H	Communication between pilot and winch operator. All Stop is required if communication is lost between pilot and winch operator. If free spooling of the winch drum occurs, the operator must go dead stick (neutral) on the winch pay in/out joystick, this will engage the internal brake system. In the event the drum continues to move the operator <i>must immediately engage the E-Stop on the control panel</i> , this will activate the external brake system. Winch Operator must be aware of Emergency Egress points if Internal Brake system fails. All Personnel will clear from work area and crew shall	2	3	M

								cease all activities. Supervisor and or OCM will go through chain of command for further instructions. **Note: The winch HPU is NOT to be powered back on until clearance from shore base is received. **			
17		Once ROV has reached depth, de energize Winch HPU	Damage to equipment, electrical shock	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Winch operator will either de energize system via control station or at the winch HPU.	1	2	L
18		Ensure that <i>cooling water has been shut off. If system has one.</i>	Waste of water and possible flooding of LARS pit. Burst hose, slips and falls.	Environmental Pollution / Damage to Equipment	2	2	M	Ensure that cooling water has been turned off at main source.	1	1	L
ROV RECOVERY											
19		ROV at depth, check weather condition prior to recovering ROV to deck.	Damage to equipment from weather patterns	Damage to Equipment	2	2	M	Communication with bridge personnel prior to exiting water	1	1	L
20		Ensure <i>cooling water pump is on. If system has one. If HPU contains air cooled system, ensure that fan is operational.</i>	Damage to HPU from overheating	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Launch/Observer confirms cooling water is on and flowing with visual of water going overboard.	2	2	M
21		Ensure that the <i>umbilical wash down is on</i> if system has been set up with on.	Damage to equipment	Damage to Equipment	2	2	M	Launch/Observer confirms that wash down water is on and flowing with visual of water on umbilical.	1	1	L

22		Energize Winch HPU.	Damage to equipment.	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	M	Winch operator obtains visual of lights and listens to HPU upon energizing system and obtain visual of all gauges and ensure that system is at proper operating pressures before leaving HPU and check all gauges. <i>Ensure that latches are open before ascending</i>	1	1	L
23		Recover the ROV following the site-specific recovery procedures.	Damage to equipment from failure to follow procedures	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Ensure personnel operating recovery system is well trained and understands the system.	2	2	M

24		Constant communication between pilot and winch operator during descent.	Possible problems with ROV during descent, hitting seabed or structures. Failure of internal winch components, Winch Internal brake system and or Control pay in/out joystick.	Personnel Injury / Environmental Pollution / Damage to Equipment	4	5	H	Communication between pilot and winch operator. All Stop is required if communication is lost between pilot and winch operator. If free spooling of the winch drum occurs, the operator must go dead stick (neutral) on the winch pay in/out joystick, this will engage the internal brake system. In the event the drum continues to move the operator <i>must immediately engage the E-Stop on the control panel</i>, this will activate the external brake system. Winch Operator must be aware of Emergency Egress points if Internal Brake system fails. All Personnel will clear from work area and crew shall cease all activities. Supervisor and or OCM will go through chain of command for further instructions. **Note: The winch HPU is NOT to be powered back on until clearance from shore base is received. **	2	3	M
25		Halt ROV at -100fsw to ensure that conditions are favourable for ROV recovery. Ensure that tension has been switched to LOW Tension and latches have been closed	Halt ROV at -100fsw to ensure that conditions are favourable for ROV recovery. Ensure that tension has been switched to LOW Tension and latches have been closed	Environmental Pollution / Damage to Equipment	3	4	M	Communication with bridge personnel prior to exiting water. Ensure that docking head is in the Drive position during recovery and latches are closed	2	2	M
26		ROV coming through splash zone.	Uncontrolled movement of ROV due to wave patterns, vessel movement, height	Environmental Pollution / Damage to Equipment	3	4	M	Timing of wave patterns prior to exit, ensure vessel is on best recovery heading possible	2	2	M

27		Latch ROV into Docking Head	Damage to equipment due to overpull.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Observe latch indicators and slowly lower ROV into docking head latches. Switch from Drive to Float and relieve tension from umbilical once latched into docking head.	2	2	M
28		Opening of LARS gates. Observer is to be 100% tied off with a full body harness with floatation.	Falling overboard, injury to personnel	Personnel Injury	2	2	L	100% tie off to prevent the hazard of falling overboard, and or injury. <i>Crew member is NOT to open Gates until ROV has been Latched in to Docking Head and tension on Umbilical has been relieved.</i>	1	1	L
29		Land ROV on to stands/deck.	Damage to equipment due to crushing ROV on stands or deck.	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	L	Constant communication with observer and winch operator.	1	1	L
30		De energize Winch HPU	Damage to equipment, electrical shock	Personnel Injury / Environmental Pollution / Damage to Equipment	2	2	L	Winch operator will either de energize system via control station or at the winch HPU.	1	1	L
31		Closing LARS gates if gates have been opened.	. Falling overboard, injury to personnel, pinch points.	Personnel Injury	2	2	L	Crew member must have full body harness with floatation and 100% tied off. <i>Crew member will not close gates if ROV has not been landed securely on deck.</i>	1	1	L
32		Place ground strap on ROV and strap ROV to deck with approved cargo straps	Electrical shock, damage to equipment, straps under tension	Personnel Injury / Damage to Equipment	2	2	L	Crew member is to install grounding strap to ROV and ensure that grounding strap is attached to vessel approved grounding area.	1	1	L
33		Personnel are to strap down ROV with approved cargo straps.	Damage to equipment, straps under tension	Personnel Injury / Damage to Equipment	2	2	L	Crew to check all straps are in good working order before use. If any are found torn, frayed, or cut these are to be removed	1	1	L

								from service and Supervisor will be informed.			
34		Ensure that cooling water and umbilical wash down has been shut off. If system has one.	Waste of water and possible flooding of LARS pit. Burst hose, slips and falls.	Damage to Equipment	2	2	L	Ensure that cooling water and umbilical wash down have been turned off at main source.	1	1	L
TRA Approval		Name		Position		Signature		Date			
By signing, the Approver confirms that the TRA has been reviewed and considers the task acceptable to proceed with residual risk meeting the acceptance criteria of DO Risk Matrix.											

*Reference DO-HSE-FRM-140 for DeepOcean Risk Assessment Matrix (RAM) and Guidewords